Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec:Sr.Super60_NUCLEUS BT | Paper -2(Adv-2024-P2-Model) | Date: 21-09-2025

Time: 02.00Pm to 05.00Pm | GTA-02 | Max. Marks: 180

KEY SHEET

MATHEMATICS

1	**B**	2	**D**	3	**D**	4	**C**	5	**BC**	6	**BCD**
7	**ABC**	8	**1**	9	**2**	10	**3**	11	**3**	12	**18**
13	**1**	14	**6**	15	**2**	16	**3**	17	**2**		

PHYSICS

18	**A**	19	**C**	20	**B**	21	**C**	22	**ABC**	23	**AD**
24	**BD**	25	**5**	26	**3**	27	**5**	28	**4**	29	**21**
30	**42**	31	**10**	32	**8**	33	**6**	34	**150**		

CHEMISTRY

35	**C**	36	**B**	37	**A**	38	**D**	39	**BD**	40	**AB**
41	**B**	42	**14**	43	**5**	44	**6**	45	**15**	46	**6**
47	**8**	48	**0.41-0.45**	49	**45.5-46.5**	50	**0.02**	51	**20**		

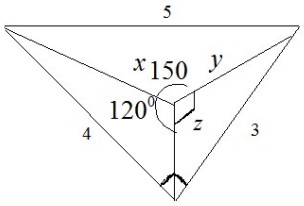
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SOLUTIONS

MATHEMATICS

1. Given $x^2+y^2-2xy\cos\frac{5\pi}{6}=25;$

$x^2+z^2-2xz\cos\frac{2\pi}{3}=16$ *and* $y^2+z^2=9$



2. Let $A=\begin{bmatrix}a & b\\ c & d\end{bmatrix}$ *and* $A+|A|=K$

$\left|\begin{bmatrix}a & b\\ c & d\end{bmatrix}\right|+\begin{bmatrix}kd & -kb\\ -kc & ka\end{bmatrix}=0$ $\qquad$ $\begin{vmatrix}a+kd & b(1-k)\\ c(1-k) & d+ka\end{vmatrix}=0$

$ad+k\left(d^2+a^2\right)+k^2ad-bc(1-k)^2=0$

$ad+k\left(d^2+a^2\right)+k^2(ad-bc)-bc+2bck=0$

$k^3+k+k\left(d^2+a^2\right)+2bck=0$

$k^3+k+k\left[\left(d^2+a^2\right)-2ad\right]+2bck=0$

$k^3+k+k(a+b)^2-2k(bc-ad)=0$

$k^3-2k^2+k+k(a+d)^2=0$

$(k-1)^2+(a+d)^2=0\Rightarrow k=1, a=-d$

$\left|\begin{bmatrix}a & b\\ c & d\end{bmatrix}-\begin{bmatrix}d & -b\\ -c & a\end{bmatrix}\right|$

$\begin{vmatrix}a-d & 2b\\ 2 & c & d-a\end{vmatrix}\Rightarrow-(a-d)-4bc$

$-a^2-d^2+2ad-4bc-(a+d)^2+4(ad-bc)$

$0+4(1)=4$

3. $g_1(x)=(x-1)^1(x-2)^2(x-2)^3....(x-19)^{19}$

$g_1(x)=(x-1)^1(x-2)^2(x-2)^3....(x-19)^{19}$

$\alpha_1=\{19,15,11,7,3\}=5$

$\beta_1=\{17,13,9,5,1\}=5$

$g_1(x)=98\times50(x-1)^{49}-100\times49(x-1)^{48}$

$=4900(x-1)^{48}(x-1)=4900(x-1)^{48}(x-2)$

$\alpha_1=1, \beta_1=0$

4. $\frac{P(x)}{x(x-1)(x+1)}=a_{97}x^{97}+a_{96}x^{96}+.....a_1x+a_0+\frac{px^2+qx+r}{x(x-1)(x+1)}$

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$$\frac{d^{100}}{dx^{100}}\left(\frac{P(x)}{x(x-1)(x+1)}\right)=\frac{d^{100}}{dx^{100}}\left(\frac{px^2+qx+r}{x(x-1)(x+1)}\right)$$

$$=-100!\frac{p(0)}{x^{100}}+\frac{100!\,p(1)}{2(x-1)^{101}}+\frac{100!\,p(-1)}{2(x+1)^{101}}$$

$$=\frac{100!}{2}\left[\frac{-2P(0)(x+1)^{101}(x-1)^{101}+P(1)x^{101}(x+1)^{101}+P(-1)x^{101}(x-1)^{101}}{x^{101}(x-1)^{101}(x+1)^{101}}\right]$$

Coefficient of x^{202} $innumerator = P(1)+P(-1)-2P(0)$

Coefficient of in numerator $= P(1)-P(-1)\neq 0$

Hence, n = 201

5. The roots of 2nd equation are a and b.

Put $x=a$ *and* $x=b$ in 1st equation and the values of p are a^2+b & a(1 + b)

6. The value is 0 by expressing 9 as product of 4 distinct integers, 3, -3, 1 and -1

7. $|\bar{x}|=|\bar{y}|=|\bar{z}|=\sqrt{2}$ *and* $\bar{x},\bar{y},\bar{z}$ *make angle of of* 60^0 *with each other*

$$\therefore \bar{x},\bar{y}=|\bar{x}|=|\bar{y}|\cos 60^0=\sqrt{2}\left(\sqrt{2}\right).\frac{1}{2}=1$$

$$\bar{y},\bar{z}=|\bar{y}|=|\bar{z}|\cos 60^0=\sqrt{2}\left(\sqrt{2}\right).\frac{1}{2}=1$$

$$and\ \bar{x},\bar{z}=|\bar{x}|=|\bar{z}|\cos 60^0=\sqrt{2}\left(\sqrt{2}\right).\frac{1}{2}=1$$

$$\bar{x},\bar{x}=|\bar{x}|^2=2$$

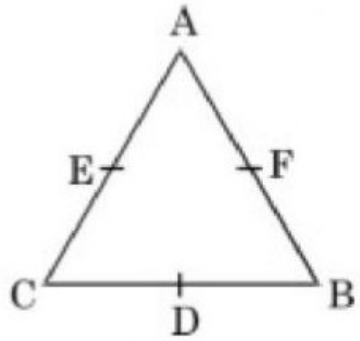
$$\bar{y},\bar{y}=|\bar{y}|^2=2\ and\ \bar{z},\bar{z}=|\bar{z}|^2=2$$

$$now,\bar{x}\times\left(\bar{y}\times\bar{z}\right)=\bar{a}\ and\ \bar{y}\times\left(\bar{z}\times\bar{x}\right)=\bar{b}$$

8. $f'(x)=1, f(x)=x$

$g(x)=1$

9.



$D(a\lambda,0,0)$

$E(0,b\lambda,0)$

$F(0,0,c\lambda)$

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$P = 2EF \Rightarrow p^2 = 4(EF)^2$

$\Rightarrow p^2 = 4\left(b^2 + c^2\right)\lambda^2$

$q^2 = 4\left(a^2 + c^2\right)\lambda^2$

$r^2 = 4\left(a^2 + b^2\right)\lambda^2$

$p^2 + q^2 + r^2 = 8\lambda\left(a^2 + b^2 + c^2\right)$

$p^2 + q^2 + r^2 = 8\lambda\left(\frac{p^2 + q^2 + r^2}{2}\right) \Rightarrow = 1/4$

$f(x) > g(x)$

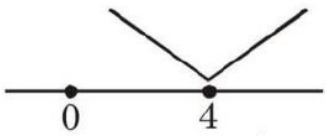
10. $|x| + \frac{4}{|x|} > p|x| + 2 \forall x \in R - \{0\}$

$p < \frac{|x| + \frac{4}{|x|} - 2}{|x|}$

$p < 1 + \frac{4}{(x)^2} - \frac{2}{|x|} \forall x \in R - \{0\}$

$Consider\ h(x) = 1 + \frac{4}{x^2} - \frac{2}{x}, x > 0$

$h'(x) = \frac{2(x-4)}{x^3}$



Now, h (4) = 1 + 1/4 - 1/2 = 3/4

p < h(x)

$\therefore 0 < p < \frac{3}{4}$

$a = 0\ b = 3/4$

$\therefore a + 4b = 3$

11. $adj(adjA) = |A|A\ and\ |adjA| = |A|$

So, if order is odd $\Rightarrow$ two possible values for A two possibilities for A (i.e. A and -A) and $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ k & \lambda & \mu \end{bmatrix}$ all have adjoints as null matrix.

12. Given, $S_k(x) = C_k x + k\int_0^1 S_{k-1}(t)dt$,

$Put\ k = 2\ and\ x = 3$

$S_2(3) = C_2 x + 2\int_0^1 S_{3-1}(t)dt$

Also,

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$$S_k(x) = C_k x + k\int_0^1 S_0(t)dt = C_1 x + \frac{x^2}{2}$$

$$S_2(3) = 3C_2 + 2\left(C_1 + \frac{t^2}{2}\right)dt = 3C_1 + 9C_1 + 9$$

Also,

$$C_1 = 1 - \int_0^1 S_0(t)dx = \frac{1}{2}$$

$$C_2 = 1 - \int_0^1 S_1(t)dx = 0$$

$$C_2 = 1 - \int_0^1 S_2(t)dx$$

$$C_3 = 1 - \int_0^1 \left(C_1 x + C_1 x^2 + \frac{x^3 3}{}\right)dx$$

$$S_2(x) = C_2 x1 + 2k\int_0^1 S_1(t)dt, where$$

$$= C_2 x + C_1 x^2 + \frac{x^3}{3}$$

$$\Rightarrow S_2(3) + 6C_3 = +6C_3 + 3C_2 + 9C_1 + 9 = 18$$

13. $$\sum_{k=1}^{\infty}\frac{(-1)^{k-1}}{k}\sum_{n=0}^{\infty}\int_0^1 x^{k2^n}dx$$

$$\int_0^1 \sum_{n=0}^{\infty}\sum_{k=1}^{\infty}\frac{(-1)^{k-1}}{k}\left(x^{2^n}\right)^k dx$$

$$\int_0^1 \sum_{2=1}^{\infty}\ln\left(1+x^{2^n}\right)$$

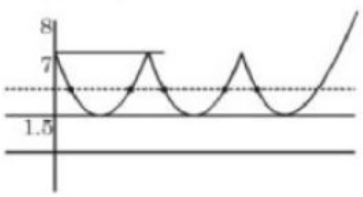
$$\int_0^1 \ln\left[(1+x)(1+x^2)(1+x^4)(1+x^8).....\right]dx$$

$$\int_0^1 \ln\left[1+x+x^2.....\infty\right]dx$$

$$\int_0^1 \ln\left(\frac{1}{1-x}\right)dx = -\int_0^1 \ln(1-x)dx = -\int_0^1 \ln x\,dx$$

$$-\left[x\ln x - x\right]^1 = -\left[\{0-1\} - \{0-0\}\right] = 1$$

14.



Min. value of $g(1) = -\frac{11}{2}$

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$f_1(x)$ has 2 non differentiable points

$f_4(x)$ has 4 non differentiable points

$f_2(x) = \|g(x)|-7|= \|6x^2-18x+8|-7|$ *for* $f_3(x)$

to have non-differentiable points

$P_2 \in (1.5, 7)$

$5 < P_1 < P_2 < P_3 < P_4 < \ldots$

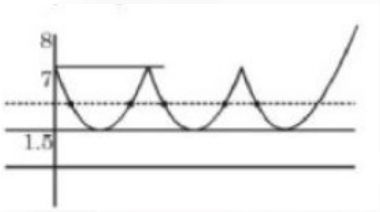
$a_1 = 2, a_2 = 4, a_3 = 6$

$$H(x) = \frac{(x-2)(x-6)}{(x-4)} = y$$

$\Rightarrow x^2 - x(8+y) + 4y + 12 = 0$

$Q(x) \geq= 0 \Rightarrow y^2 + 16 \geq 0 \Rightarrow y \in R$

15.



Min. value of $g(1) = -\frac{11}{2}$

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$$H(x) = \frac{(x-2)(x-6)}{(x-4)} = y$$

$\Rightarrow x^2 - x(8+y) + 4y + 12 = 0$

$Q(x) \geq= 0 \Rightarrow y^2 + 16 \geq 0 \Rightarrow y \in R$

16. $$A = \int_0^{1/2} x^2 dx = \frac{1}{3}\left[x^2\right]_0^{1/2} = \frac{1}{24}$$

$$\frac{1}{24} = \frac{1}{3}\left[\left[x^2\right]_\beta^{1/2} + \left[(x-1)^3\right]_{1/2}^{\beta+f(\beta)}\right]$$

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$$\frac{1}{8}=\frac{1}{8}-\beta^3+\left(f(\beta)-1\right)^3-\left(-\frac{1}{2}\right)^3$$

$$\beta^3=\left(\beta+f(\beta)-1\right)^3+\frac{1}{8}$$

$$f(\beta)=(1-\beta)+\left(\beta^2-1/8\right)^{1/3}$$

$$f(x)=(1-x)+\left(x^3-1/8\right)^{1/3}$$

$$\lim_{x\to 1/2}\frac{\left(f(x)+x-1\right)^3}{\left(x-\frac{1}{2}\right)}=\lim_{x\to 1/2}\frac{\left(x^3-1/8\right)}{-1/2}=\lim_{x\to 1/2}\left(x^2+\frac{1}{2}x+\frac{1}{4}\right)$$

$$=\frac{1}{4}+\frac{1}{4}+\frac{1}{4}=\frac{3}{4}$$

$$f0-1+\frac{1}{3}\left(x^3-\frac{1}{8}\right)^{-2/3}\left(3x^2\right)$$

$$\left(x^3-\frac{1}{8}\right)^{-2/3}=x^2\Rightarrow\left(x^3-\frac{1}{8}\right)^2=x^6$$

$$so\ x^3-\frac{1}{8}=-x^2\Rightarrow 2x^3=\frac{1}{8}$$

$$so\ x^3=\frac{1}{16}$$

17. $$A=\int_0^{1/2}x^2dx=\frac{1}{3}\left[x^2\right]_0^{1/2}=\frac{1}{24}\frac{1}{24}=\frac{1}{3}\left[\left[x^2\right]_\beta^{1/2}+\left[(x-1)^3\right]^{\beta+f(\beta)}\right]$$

$$\frac{1}{8}=\frac{1}{8}-\beta^3+\left(f(\beta)-1\right)^3-\left(-\frac{1}{2}\right)^3$$

$$\beta^3=\left(\beta+f(\beta)-1\right)^3+\frac{1}{8}$$

$$f(\beta)=(1-\beta)+\left(\beta^2-1/8\right)^{1/3}$$

$$f(x)=(1-x)+\left(x^3-1/8\right)^{1/3}$$

$$\lim_{x\to 1/2}\frac{\left(f(x)+x-1\right)^3}{\left(x-\frac{1}{2}\right)}=\lim_{x\to 1/2}\frac{\left(x^3-1/8\right)}{-1/2}=\lim_{x\to 1/2}\left(x^2+\frac{1}{2}x+\frac{1}{4}\right)$$

$$=\frac{1}{4}+\frac{1}{4}+\frac{1}{4}=\frac{3}{4}$$

$$f0-1+\frac{1}{3}\left(x^3-\frac{1}{8}\right)^{-2/3}\left(3x^2\right)$$

$$\left(x^3-\frac{1}{8}\right)^{-2/3}=x^2\Rightarrow\left(x^3-\frac{1}{8}\right)^2=x^6$$

$$so\ x^3-\frac{1}{8}=-x^2\Rightarrow 2x^3=\frac{1}{8}$$

$$so\ x^3=\frac{1}{16}$$

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PHYSICS

18. Volume of ice block $= 6\times6\times6 = 216cm^3$

Volume of water $= 10\times10\times6 - 6\times6\times6\times5.4 = 405.6cm^3$

$[\because$ Length of ice block submerged $= \frac{P_{ice}}{P_{\omega}}\times6 = \frac{0.9}{1}\times6 = 5.4\ cm\ ]$

Heat rejected by water when it cools from 16.15^0C to 0^0C

$Q = ms\Delta\theta = 405.6\times1\times16.15 = 6550.4$cal

Mass of ice that melts $= \frac{6550.4}{80} = 81.88g$

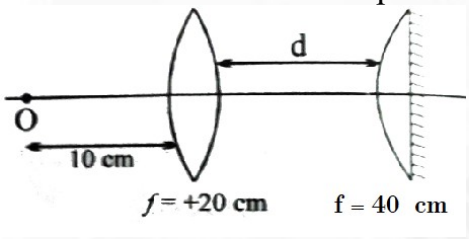
Volume of ice that melts $= \frac{81.88}{0.9} = 91cm^3$

$\therefore$ Volume of ice block in thermal equilibrium at 0^0C will be $= 6^3 - 91 = 125cm^3$.

$\therefore$ Side length = 5 cm.

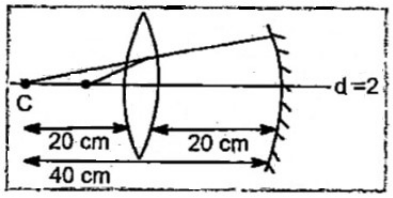
$\therefore$ Length of the submerged part $= 0.9\times5 = 4.5cm$.

19. The silvered lens can be replaced by a minor of focal length given as

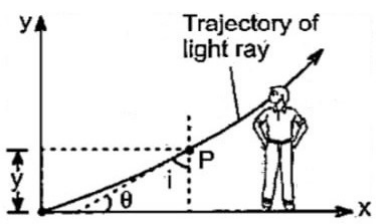


$\frac{1}{F_M} = \frac{1}{f_m} - \frac{2}{f_1}, \frac{1}{F_M} = 0 - \frac{2}{40} F_M = -20$

For lens $\upsilon = \frac{uf}{u+f}$ $\upsilon = \frac{-10\times20}{-10+20} = -20$



20. Let O be they distant object just-visible to the man. Let P be a point on the trajectory of the ray. From Fig. θ = 90-i.



The slope of tangent at point P is $tan\theta = \frac{dy}{dx} = cot\ i.$

From Snell's law, n sin i = constant.

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At the surface $n = n_0$ *and* $i = 90^o$

$$n_0 \sin 90^o = n \sin i = \left(n_0\sqrt{1+ay}\right)\sin i$$

$$\sin i = \frac{1}{\sqrt{1+ay}}$$

$$\cot i = \frac{dy}{dx} = \sqrt{ay}$$

$$\int_0^y \frac{dy}{\sqrt{ay}} = \int_0^x dx$$

$$x^1 = 2\sqrt{\frac{y}{a}}$$

On Substituting $y = 2.0\,m$ *and* $a = 2\times10^{-6}\,m^{-1}$, we have

$$x_{\max} = 2\sqrt{\frac{2}{2\times10^{-6}}} = 2000m$$

21. Let L is the original length of the wire and k is force constant of wire.

Final length initial + elongation $L' = L + \frac{F}{K}$

for first condition, $a = L + \frac{4}{k}$

For second condition, $b = L + \frac{5}{k}$

by solving Eqs. (i) and (ii) we get ltbr L = 5a - 4b and $k = \frac{1}{b-a}$

Now, when the longitudinal tension is 9 N. Length of the string

$= L + \frac{9}{k} = 5a - 4b + 9(b-a) = 5b - 4a$

22. $\overrightarrow{a_0} = \left(2g\hat{i} + g\hat{j}\right)$

$$\vec{F}_B = \rho V\left(\overrightarrow{a_0} - \vec{g}\right) = 2\rho Vg\left(\hat{i}+\hat{j}\right)$$

$$\vec{F}_{net} = \vec{F}_B + m\vec{g} = 2\rho Vg\left(\hat{i}+\hat{j}\right) - \rho_0 Vg\hat{j}$$

$$\vec{a}_{net} = \frac{2\rho}{\rho_0}g\left(\hat{i}+\hat{j}\right) - g\hat{j}$$

$$\vec{a}_{relative} = \vec{a}_{net} - \vec{a}_0$$

23. Let the displacement of the string be given by $y(x,t) = A\sin\frac{\pi x}{L}\cos(\omega t + \delta)$ where δ is a phase factor. So the transverse velocity is given by

$$y(x,t) = \frac{\partial y}{\partial t} = -\omega A\sin\frac{\pi x}{L}\sin(\omega t + \delta)$$

The maximum kinetic energy is equal to the string's total energy of oscillation. Note that all points of the string achieve their maximum kinetic energy at the same instant of time,

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where $y=0$ for all x. Since $dm=\mu dx$ where $\mu=\left(\frac{m}{L}\right)$ is the mass per unit length of the uniform string. The maximum kinetic energy,

$$E_K=\max.\,of\left[\frac{1}{2}\int\left(\frac{\partial y}{\partial t}\right)^2 dm\right]$$

$$=\max.\,of\left[\frac{1}{2}\int_0^L \mu\left(\frac{\partial y}{\partial t}\right)^2 dx\right]$$

The maximum value of $\left(\frac{\partial y}{\partial t}\right)^2$, occurs when $\sin^2(\omega t+\delta)=1$

Hence, $E_K=\frac{\mu}{2}A^2\omega^2\int_0^L \sin^2\frac{\pi x}{L}dx$

The integral $\int_0^L \sin^2\frac{\pi x}{L}dx$ over the half-cycle has the average value of $\frac{L}{2}$

24. The condition for getting maxima is $d\sin\theta=m\lambda$. The wavelength of electron will be given as $\lambda=hc/v$. The distance between successive maxima will increase if θ becomes smaller. For this either d should be increased or λ should be decreased. For the latter we must increase the voltage V.

25.

$$\frac{GM}{x^2}=\frac{GM/81}{(60R-x)^2}$$

$$\Rightarrow\left(\frac{60R-x}{x}\right)^2=\frac{1}{81}$$

$$\Rightarrow\frac{60R-x}{x}=\frac{1}{9}$$

$$\Rightarrow 54R=x$$

$$U_{\max}=-\frac{GMm}{54R}-\frac{G\frac{M}{81}}{6R}$$

$$[\because 60R-54R=6R]$$

$$=-\frac{GMm}{R}\left[\frac{1}{54}+\frac{1}{81\times 6}\right]$$

$$=-\frac{GMm}{R}\left[1+\frac{1}{9}\right]$$

$$=-\frac{5GMm}{243R}$$

26. CONCEPTUAL

27. Field at A:

Due to 'ρ' is $E_1=\frac{\rho}{3\varepsilon_0}\left(\frac{R}{2}\right)$ *parallel to* C_1C_2

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Due to '2ρ' is $E_2 = \frac{2\rho}{3\varepsilon_0}\left(\frac{R}{4}\right)$ *parallel to* $C_2C_3 = \frac{\rho}{3\varepsilon_0}\frac{R}{2}$

Due to '3ρ' is $E_3 = \frac{4\rho}{3\varepsilon_0}.\frac{R}{8}$ *parallel to* $C_3A = \frac{\rho}{3\varepsilon_0}\frac{R}{2}$

$\therefore \quad E_A = \sqrt{E_1^2 + (E_2 + E_3)^2} = \frac{\sqrt{5}}{6}\frac{\rho R}{\varepsilon_0}$

28. Power radiated by the filament

$$P = \sigma e\left(T^4 - T_0^4\right)$$

$$= \sigma e T^4 \qquad \left[\therefore T^4 >> T_0^4\right]$$

But $P = VI$

$\therefore \quad VI = \sigma e T^4$

$\therefore \quad \ln(VI) = \ln(\sigma e) + 4\ln T$

$\therefore$ *graph has a slope* $= 4$.

29.

The force by the gas on the piston is

$$F = p_0 A + kx$$

Where $\rho_0 = 100\ kPa$ is the atmospheric pressure, A = 20 cm² is the area of cross section, $k = 200\ N\,m^{-1}$ is the spring constant and x is the compression of the spring. The work done by the gas as the piston moves through $l = 10$ cm is

$$W = \int_0^l F\ dx$$

$$= p_0 Al + \frac{1}{2}kl^2$$

$$= \left(100\times10^3\,Pa\right)\times\left(20\times10^{-4}m^2\right)\times\left(10\times10^{-2}m\right)$$

$$+\frac{1}{2}\left(200\ N\,m^{-1}\right)\times\left(100\times10^{-4}m^2\right)$$

$$= 20J + 1J = 21J.$$

30. Piston A is free to move, so gas will expand isobarically and piston B is fixed, so the process is isochoric. Given that, $Q_A = Q_B$

$$\Rightarrow nC_P\Delta T_A - nC_V\Delta T_B \Rightarrow \left(\frac{7R}{2}\right)(30) = \left(\frac{5R}{2}\right)\Delta T_B \Rightarrow \Delta T_B = 42K$$

31. $\omega = \omega\sqrt{\frac{M}{M+m}}$ $\qquad x_1 = A\sin\phi_1$

Potential energy stored in the spring at $(t = t_1) = \frac{1}{2}kx_1^2 = \frac{1}{2}M\omega^2 sin^2\phi_1$

Let v and v. be the velocity of system just before collsion and just after collision, so using COLM $\upsilon = \frac{M\upsilon}{(M+m)} = \frac{MA\omega\cos\theta}{(M+m)}$

Total energy after collision

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$(t=t_1)=\frac{1}{2}kx_1^2=\frac{1}{2}M\omega^2 sin^2\phi_1$

$\upsilon=\frac{M\upsilon}{(M+m)}=\frac{MA\omega\cos\theta}{(M+m)}$

$=PE+KE=\frac{1}{2}kx_1^2+\frac{1}{2}(M+m)v^2$

$=\frac{1}{2}MA\omega^2\left[\frac{M+m\sin^2\theta}{M+m}\right]$

$=\frac{1}{2}kA^2\left[\frac{M+m\sin^2\theta}{M+m}\right]=8J$

$\frac{1}{2}(M+m)A^2\omega^2=\frac{1}{2}MA^2\omega^2\left[\frac{M+m\sin^2\theta}{M+m}\right]$

$A=A\sqrt{\frac{M+m\sin^2\theta}{M+m}}=8cm$

32. $\omega=\omega\sqrt{\frac{M}{M+m}}$

$x_1=A\sin\phi_1$

Potential energy stored in the spring at $(t=t_1)=\frac{1}{2}kx_1^2=\frac{1}{2}M\omega^2 sin^2\phi_1$

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$(t=t_1)=\frac{1}{2}kx_1^2=\frac{1}{2}M\omega^2 sin^2\phi_1$

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$A=A\sqrt{\frac{M+m\sin^2\theta}{M+m}}=8cm$

33. $R_{ab}\ R_{cd}=R_{ad}\ R_{ba}$

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$$\frac{pl}{A}\frac{L_{cd}}{A}=\frac{P(2l)}{A}\frac{P(3l)}{A}$$

$$l_{cd}=6l$$

$$\frac{T_A-T_B}{R_{AB}}=\frac{T_B-T_C}{R_{BC}}$$

34. $R_{ab}\ R_{cd}=R_{ad}\ R_{ba}$

$$\frac{pl}{A}\frac{L_{cd}}{A}=\frac{P(2l)}{A}\frac{P(3l)}{A}$$

$$l_{cd}=6l$$

$$\frac{T_A-T_B}{R_{AB}}=\frac{T_B-T_C}{R_{BC}}$$

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CHEMISTRY

35. $P, Q, R \rightarrow NGP$

36. $\Delta H_{323} = \Delta H_{373} + \left(\Delta C_p\right)_r [323-373]$

$$\Delta S_{323} = \Delta S_{373} + \left(\Delta C_p\right)_r ln\frac{323}{373}$$

37. $\Delta 7_b = K_b \times m$

$$K_b = \frac{R7_b^2 \times M}{1000\ \Delta\ H_{vap}}$$

$$\Delta S_{vap} = \frac{\Delta H_{vap}}{B.P}$$

38. $N_2O_5 \rightarrow N_2O_5 + O_2$

$y-x \quad x-y \quad x+y$

$N_2O_3 \rightarrow N_2O + O_2$

$x-y \qquad y \qquad x+y$

$$\frac{(x-y)(2.5)}{4-x} = 2.5$$

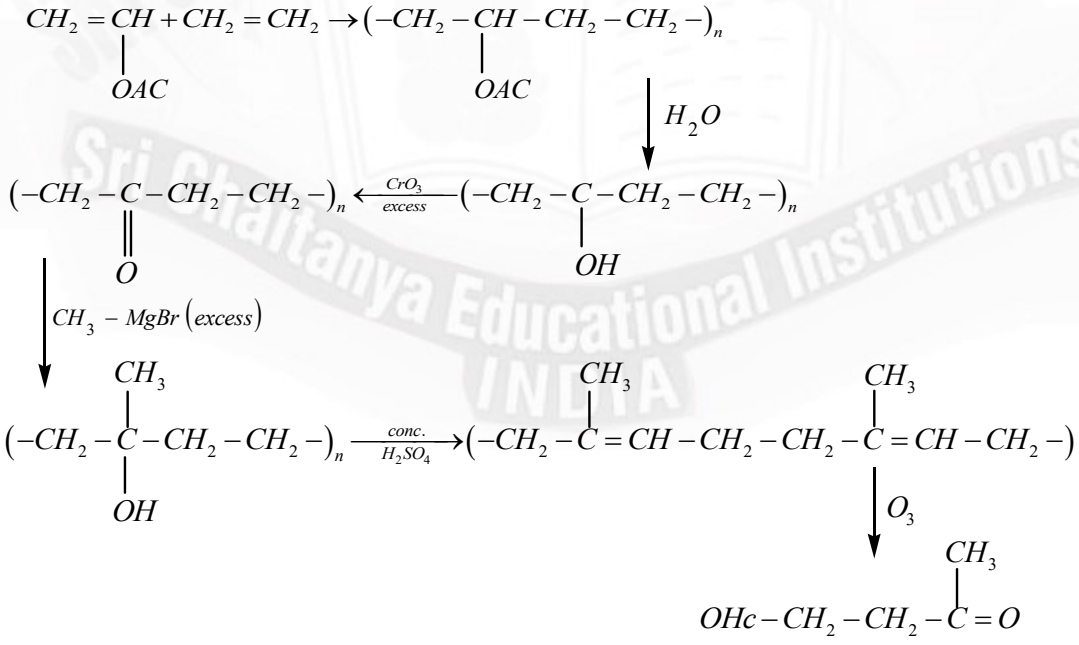
$$x-y = 2.5$$

39.

$\left[Co(H_2O)_6\right]^{3+}, \left[Co(NH_3)_6\right]^{3+}, \left[Co(NO_2)_6\right]^{3-}, \left[NiF_6\right]^{2-}, \left[Mn(CN)_6\right]^{3-}, \left[Fe(bpy)_3\right]^{3+}, \left[Fe(phen)_3\right]^{3+},$
$\left[Fe(en)_3\right]^{3+}, \left[Fe(phen)_3\right]^{2+}, \left[Fe(bpy)_3\right]^{2+}, \left[CoF_6\right]^{2-}, \left[Cr(CN)_6\right]^{4-}, \left[RuCl_6\right]^{2-}, \left[Co(NH_3)_4(NO_3)_2\right]^{+}$ l ow pin.

40. $\frac{432}{96.4} = \frac{1242}{\lambda(nm)} \qquad \lambda \cong 277$

41. Conceptual



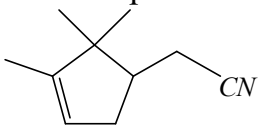
42.

43. a,b,f,g,h

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44. Except F & H

45. Conceptual

46. 

47. $Y = O$

$X = N$

$Z = F$

48. Conceptual

49. Conceptual

50. $IO_3^- + 5I^- + 6H^+ \rightarrow 3I_2 + 3H_2O$

$I_2 + 2_2^5O_3^{2-} \rightarrow 5_4O_6^{2-} + 2I^-$

$0.02 \times 1 \times 24 = mill\ mole\ of\ I_2 \times 2$

$0.01 \times 24 \times \frac{5}{3} = M_{HCl6} \times 24 \times 5$

$M = 0.02$

51. $IO_3^- + 5I^- + 6H^+ \rightarrow 3I_2 + 3H_2O$

$I_2 + 2_2^5O_3^{2-} \rightarrow 5_4O_6^{2-} + 2I^-$

$0.02 \times 1 \times 24 = mill\ mole\ of\ I_2 \times 2$

$0.01 \times 24 \times \frac{5}{3} = M_{HCl6} \times 24 \times 5$

$M = 0.02$

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